

Chapter 23: LangGraph Interrupts, Human Approval, Resume Semantics, and Advanced Human-in-the-Loop Architectures

Part III: LangGraph Deep Dive

Book: Agentic AI & RAG Systems Handbook

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Chapter Overview

Most AI workflows assume execution is continuous:

1	Start
2	↓
3	Execute
4	↓
5	Finish

Enterprise workflows rarely work this way.

Real-world systems frequently require pauses:

- Human approvals
- Regulatory reviews
- Customer responses
- External system confirmations
- Escalations
- Risk assessments

Traditional workflow engines struggle with pauses that may last:

1	Minutes
2	
3	Hours
4	
5	Days
6	
7	Weeks

LangGraph introduces **Interrupts** as a first-class execution pattern.

Interrupts allow an agent to:

1	Pause
2	↓
3	Wait
4	↓
5	Resume

without losing workflow state.

This capability is foundational to:

- Human-in-the-Loop systems
 - Enterprise governance
 - Long-running agents
 - Regulatory workflows
 - Production operations platforms
-

Learning Objectives

After completing this chapter you should be able to:

- Understand interrupt architecture
 - Implement resumable workflows
 - Design approval-driven agents
 - Build enterprise HITL systems
 - Integrate interrupts with checkpointing
 - Design approval governance workflows
 - Manage long-running human decisions
 - Build LangGraph + Temporal architectures
 - Explain resume semantics during interviews
-

Prerequisites

Required:

- Chapter 9: Human-in-the-Loop Pattern
- Chapter 18: State Graph Fundamentals
- Chapter 22: Checkpointing and Durable Execution

Recommended:

- Workflow orchestration knowledge
 - Enterprise governance experience
-

1. Introduction

Imagine a deployment workflow:

```
1 Analyze Release
2     ↓
3 Generate Recommendation
4     ↓
5 Human Approval
6     ↓
7 Deploy
```

Approval may take:

```
1 1 Hour
2
3 1 Day
4
5 3 Days
```

The workflow must wait.

A normal application may fail because:

```
1 Process Restart
2
3 Server Reboot
4
5 Deployment
6
7 Node Failure
```

Interrupts solve this problem.

Instead of remaining active:

```
1 Workflow
2     ↓
3 Pause
4     ↓
```

5	Save State
6	↓
7	Resume Later

2. Core Concepts

What Is An Interrupt?

Definition

An interrupt is a controlled pause in graph execution that allows the workflow to stop and later resume from the same point.

Core Principle

1	Execute
2	↓
3	Interrupt
4	↓
5	Persist
6	↓
7	Wait
8	↓
9	Resume

Why Interrupts Matter

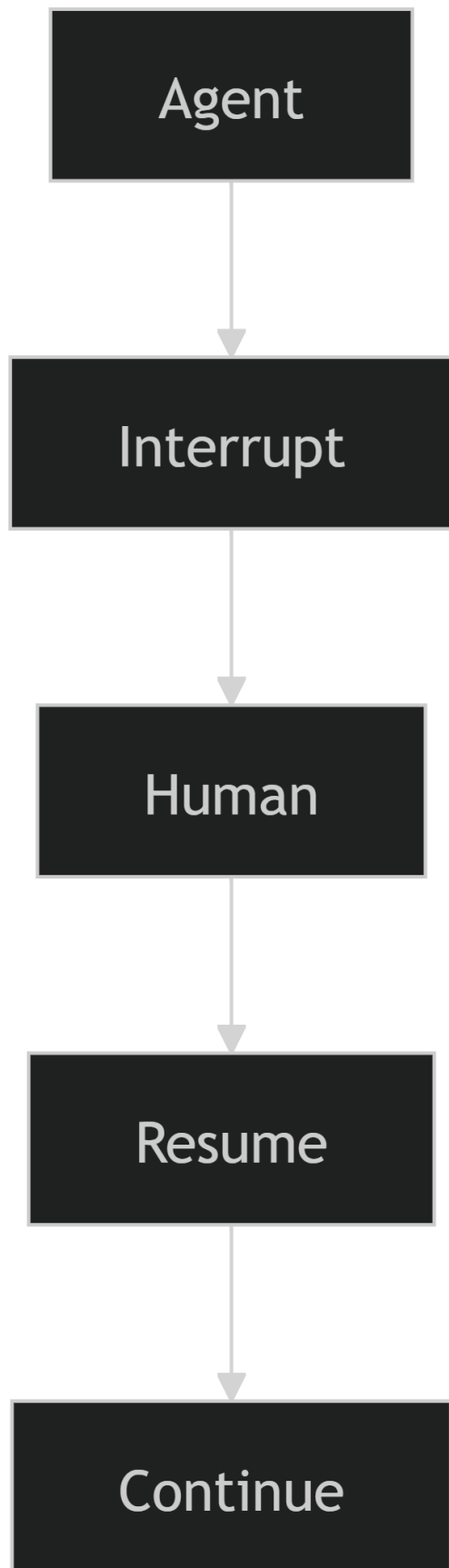
They enable:

- Human approvals
- External event waiting
- Regulatory reviews
- Long-running processes

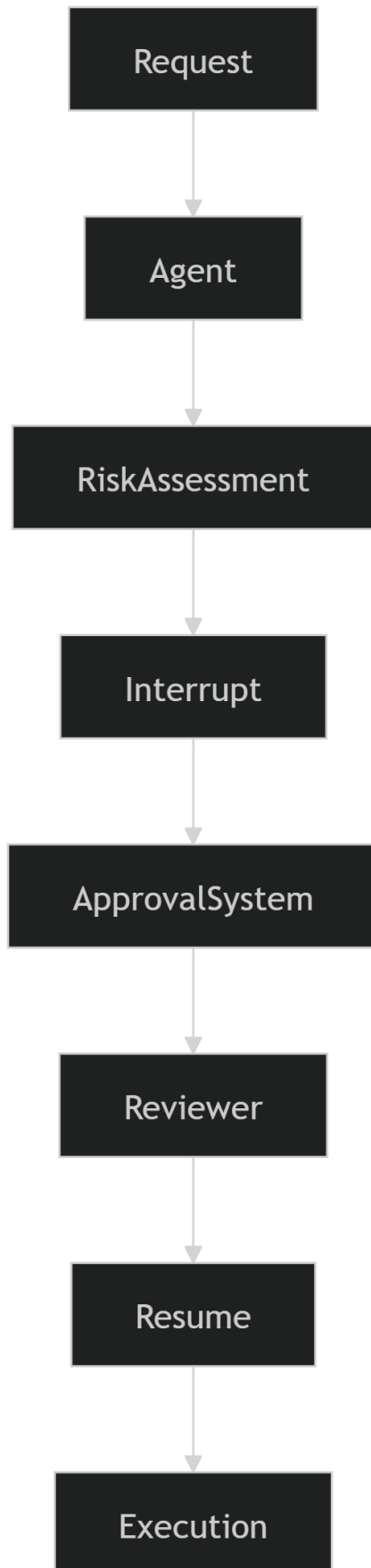
without losing state.

3. Interrupt Architecture

Basic Architecture



Enterprise Architecture



4. Human Approval Example

Request:

1 Deploy To Production

Workflow:

1	Analyze
2	↓
3	Evaluate Risk
4	↓
5	Interrupt
6	↓
7	Await Approval
8	↓
9	Resume
10	↓
11	Deploy

The workflow can safely pause.

5. Interrupt Lifecycle

Stage 1

Workflow Executes

Stage 2

Interrupt Triggered

Stage 3

State Persisted

Stage 4

Workflow Suspended

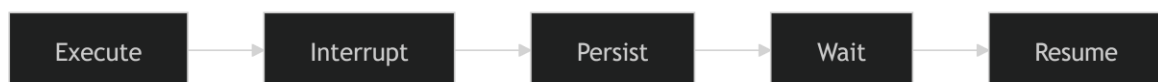
Stage 5

External Input Received

Stage 6

Workflow Resumes

Architecture



6. LangGraph Interrupt API

Conceptually:

```
1 from langgraph.types import interrupt
```

Example

```
1 def approval_node(state):
2
3     decision = interrupt(
4         {
5             "request":
6                 "Approve deployment?"
7         }
8     )
9
10    return {
11        "approved": decision
12    }
```

Execution stops until input arrives.

7. Resume Semantics

Definition

Resume semantics define how execution continues after an interrupt.

Core Principle

Resume from:

1	Saved State
---	-------------

not from:

1	Beginning
---	-----------

Architecture

Checkpoint



Resume



NextNode

8. Why Resume Semantics Matter

Incorrect resumes create:

- Duplicate actions
- State corruption
- Lost approvals
- Compliance failures

Example:

```
1 Ticket Already Created
```

Workflow must not repeat the action.

9. Resume Object Design

Typical resume payload:

```
1 {  
2   "approved": true,  
3   "approver": "manager",  
4   "comments": "approved"  
5 }
```

Contains:

- Decision
 - Actor
 - Metadata
 - Timestamps
-

10. Approval Workflow Design

Simple Approval

Agent



Approval



Execute

Multi-Step Approval

Agent



ManagerApproval



SecurityApproval



Execute

Enterprise Approval

Agent



Manager



Compliance



Security



Execute

11. Approval SLA Management

Approvals must not wait indefinitely.

Example

1	<code>approval_timeout:</code>
2	<code>24h</code>

Timeout Actions

Auto-Reject

1	No Approval
---	-------------

Escalate

1	Escalation Queue
---	------------------

Follow-Up

1	Reminder
---	----------

12. Interrupts Beyond Human Approval

Interrupts are not limited to HITL.

External System Waits

Example:

1	Payment Confirmation
---	----------------------

Customer Response

Example:

1 Need Additional Documents

Event Arrival

Example:

1 File Upload

Compliance Reviews

Example:

1 Audit Approval

13. Waiting For External Events

Architecture:

Workflow



Interrupt



Event



Resume



Continue

This enables event-driven recovery.

14. Interrupts and Checkpointing

Interrupts depend on persistence.

Without checkpointing:

1	Interrupt
2	↓
3	Crash
4	↓
5	Lost State

With checkpointing:

1	Interrupt
2	↓
3	Persist
4	↓
5	Crash
6	↓
7	Resume

Architecture

Interrupt



Checkpoint



Storage



Resume

15. Interrupts and Temporal

One of the strongest enterprise combinations.

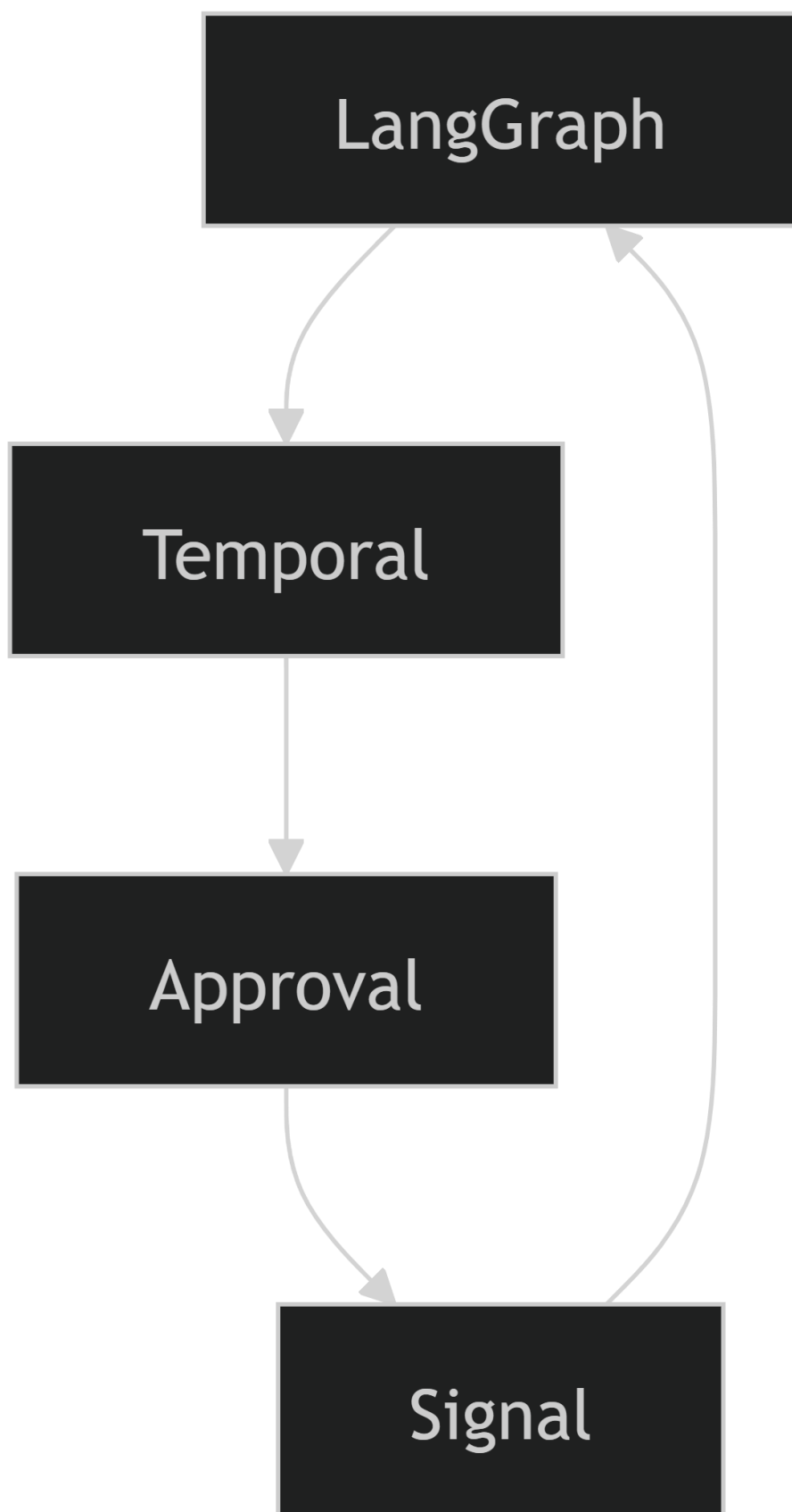
LangGraph Responsibilities

- | | |
|---|-----------|
| 1 | Reasoning |
| 2 | |
| 3 | State |
| 4 | |
| 5 | Routing |

Temporal Responsibilities

- | | |
|---|------------|
| 1 | Durability |
| 2 | |
| 3 | Waiting |
| 4 | |
| 5 | Signals |
| 6 | |
| 7 | Scheduling |

Combined Architecture



16. DevOps Approval Example

Workflow:

1	Deployment Request
2	↓
3	Analysis
4	↓
5	Interrupt
6	↓
7	Approval
8	↓
9	Deployment

Benefits:

- Traceable
 - Auditable
 - Resilient
-

17. Financial Workflow Example

Request:

1	Transfer Funds
---	----------------

Process:

1	Risk Review
2	↓
3	Approval
4	↓
5	Transfer

Interrupt protects execution.

18. Compliance Workflow Example

Request:

1	Policy Exception
---	------------------

Workflow:

1	Assessment
2	↓
3	Interrupt
4	↓
5	Compliance Officer
6	↓
7	Resume

19. Governance Architecture

Interrupts become governance control points.

Agent



Interrupt



Governance



Resume

Questions:

- 1 Who Approved?
- 2
- 3 Why?
- 4
- 5 When?
- 6
- 7 What Changed?

20. Auditability

Capture:

- Approval ID
- Reviewer
- Timestamp
- Decision
- Evidence
- Workflow ID

Example Audit Record

```
1 {  
2   "workflow_id": "123",  
3   "action": "deploy",  
4   "approved": true,  
5   "reviewer": "security_lead"  
6 }
```

21. Security Considerations

Approval systems require:

Authentication

Who approved?

Authorization

Were they allowed?

Integrity

Can approval records be trusted?

Non-Repudiation

Can actions be traced?

22. Observability

Track:

Active Interrupts

Waiting workflows.

Approval Latency

Time to decision.

Timeout Rate

Expired approvals.

Resume Success Rate

Recovery quality.

Escalation Rate

Workflow health.

23. Scalability

Large organizations may have:

1 Thousands

of paused workflows simultaneously.

Requirements:

- Durable storage
 - Efficient retrieval
 - Approval indexing
 - Searchability
-

24. Failure Modes

Failure 1

Resume With Wrong State

Failure 2

Approval Lost

Failure 3

Expired Interrupt

Failure 4

Duplicate Resume

Failure 5

Orphaned Workflow

No reviewer responds.

25. Common Mistakes

Mistake 1

No timeout handling.

Mistake 2

No audit trail.

Mistake 3

Using interrupts for everything.

Mistake 4

Lack of checkpointing.

Mistake 5

Resume logic not tested.

26. Anti-Patterns

Human Approval For Every Action

Creates bottlenecks.

No Approval For High-Risk Actions

Creates risk.

Hidden Interrupts

Poor governance.

Restart Instead of Resume

Loses progress.

27. Trade-Offs

Benefit	Cost
Governance	More complexity
Human oversight	Added latency
Durability	Storage overhead
Auditability	More infrastructure

28. Architecture Decision Matrix

Scenario	Interrupts
HITL Workflows	Essential
Compliance	Essential
Financial Systems	Essential

Deployments	Strongly Recommended
Investigations	Strong
FAQ Bot	Not Needed

29. Real-World Use Cases

DevOps Deployments

Production approvals.

Compliance Reviews

Exception handling.

Financial Transactions

Authorization workflows.

Healthcare

Clinical review processes.

Agentic Operations Platforms

Escalation workflows.

30. Interview Questions

Q1

What is an interrupt?

Q2

Why are interrupts important?

Q3

How do interrupts differ from checkpoints?

Q4

What are resume semantics?

Q5

How do LangGraph and Temporal work together?

Q6

What is the biggest interrupt failure?

Q7

How should approvals be modeled?

Q8

When should interrupts be used?

31. Interview Answers

Q1

A controlled pause in workflow execution that allows later resumption.

Q2

Because enterprise workflows often require human decisions and long wait periods.

Q3

Checkpoints persist state; interrupts pause execution.

Q4

The rules governing how workflows continue after interruption.

Q5

LangGraph manages reasoning while Temporal manages durable waiting and recovery.

Q6

Resuming with invalid or outdated state.

Q7

Using explicit approval records, audit metadata, and governance controls.

Q8

For human approvals, external waits, escalations, and long-running workflows.

32. Architecture Review Checklist

Interrupt Readiness

- Interrupt strategy defined
 - Approval model documented
 - Resume semantics defined
 - Timeout policies configured
 - Checkpoint integration implemented
 - Audit logging enabled
 - Security controls reviewed
 - Observability configured
 - Recovery tested
-

33. Practical Exercises

1. Design a deployment approval workflow.
 2. Implement an interrupt node in LangGraph.
 3. Design resume payload structures.
 4. Model a compliance-review approval process.
 5. Build an interrupt + checkpoint architecture.
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34. Key Takeaways

- Interrupts enable resumable workflows.
 - Human approvals are a primary use case.
 - Resume semantics are critical to correctness.
 - Interrupts depend heavily on checkpointing.
 - LangGraph and Temporal complement each other naturally.
 - Governance and auditability are essential.
 - Not every action requires interruption.
 - Timeouts and escalation paths are mandatory.
 - Enterprise workflows frequently rely on long-running pauses.
 - Interrupt architecture is foundational to production-grade HITL systems.
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35. References

- [LangGraph Documentation](#)
 - [Temporal Documentation](#)
 - [Durable Execution Patterns](#)
 - [Human-in-the-Loop Design Frameworks](#)
 - [Workflow Orchestration Research](#)
 - [Enterprise Governance Architecture Patterns](#)
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Chapter Status: Completed

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Next Chapter: Chapter 24 – LangGraph Multi-Agent Architectures, Agent Orchestration, Supervisor Graphs, and Collaborative Execution Systems